# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Seventh Semester of B. Tech. (EC) Examination**

**May 2014**

**EC404 Embedded Systems**

**Date: 22.05.2014, Thursday** **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

***Instructions:***

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

## SECTION – I

| | | | |
|---|---|---|---|
| Q - 1 | (a) | Define Embedded system. List real time applications of embedded system and explain any one in detail. | [07] |
| Q - 2 | (a) | Compare features of following: a microprocessor, microcontroller, DSP and ASSP. | [07] |
| | (b) | What are the challenges faced in designing an embedded system? | [07] |
| | | **OR** | |
| | (b) | Interface EEPROM 2402 with 8051 microcontroller. Also write a 8051 program to write and read the data from 2402 using I2C protocol. | [07] |
| Q – 3 | | **Attempt any two.** | [14] |
| | (a) | List different communication buses used in embedded system. With help of neat sketch explain CAN bus. | |
| | (b) | What is RTOS? How does it differ from conventional OS? Describe three function and their activities provided by RTOS services. | |
| | (c) | With neat diagram, Interface LCD with LPC2148, also write a C program to display **"WELCOME"**. | |

## SECTION – II

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|---|---|---|---|
| Q - 4 | (a) | Differentiate ARM mode and thumb mode | [03] |
| | (b) | Differentiate general purpose processors and DSP processors. | [04] |
| Q - 5 | (a) | What is advantage and disadvantage of pipeline? With neat sketch explain 3-stage pipeline and its drawback. | [07] |
| | (b) | Tabulate hardware units needed in each of the systems: camera and cell phone. | [07] |
| | | **OR** | |
| Q - 5 | (a) | What is hardware software co-design? Explain fundamental issues in hardware software co-design. | [07] |
| | (b) | List software tools used in embedded system design. Explain the role of IDE for Embedded software development. | [07] |

Q - 6 **Attempt any two.** **[14]**

(a) With the help of example, Explain following RTOS scheduling

i) Cooperative Scheduling of Ready Tasks using Precedence Constrains,
ii) Preemptive scheduling.

(b) Explain multiprocessing, multitasking, multiprogramming, context switching ,context saving and context retrieval. What all activities are involved in context switching?

(c) Explain the different Life Cycle Models adopted in embedded product development.

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